Text mode file operations

Problem: Word Frequency

In this problem, you are given a text file called “paragraph.txt”. You need to count the word frequencies. However, we are not interested in all the words. There is another file called “keywords.txt” which lists the words of interest. In this file, all the words are put on one line. They are separated by spaces. The last word is followed by a new line character. All the keywords are mentioned in lower case. Each keyword is no more than 20 characters; and there are at most 100 keywords. For each word in this file, we would like to know, how many times does it occur in the “paragraph.txt” file? Output each keyword and its frequency in a file called “freq.txt”.

Hint: First read the keywords using the %s format-specifier of fscanf() method. Store them in the following declared array of structures:

```
struct keyword
{
      char word[20 + 1];
      int freq;
} words[100];
```

Then read each word of “paragraph.txt” using the %s format-specifier of fscanf() method. If the first letter of the string is capitalized, you need to lower its case. Also, check if the last letter is a punctuation character or not. If there is a punctuation character, remove it using the following code segment:

```
len = strlen(str);
if (ispunct(str[len - 1]))
      str[len - 1] = '\0';
```

paragraph.txt
In this problem, you are given a file containing temperature information of different cities of Bangladesh Assume that number of cities would be no more than 100. You need to perform some processing on this information and store the processed information into some other files. Read the input file; store the information of each city in an appropriately designed structure. After processing of a file is done, remember to close it.

keywords.txt
you are done when store information in file on it perform processing

freq.txt (output)
you 2 are 1 done 1 when 0 store 2 information 4 in 2 file 3 on 1 it 1 perform 1 processing 2

Problem: City Heat

In this problem, you are given a file containing temperature information of different cities of Bangladesh (Assume, number of cities would be no more than 100). You need to perform some processing on this information and store the processed information into some other files. Read the input file and store the information of each city in an appropriately designed structure. After processing of a file is done, remember to close it.

Following information is stored in the different input files:

- **cities.txt**: Each line contains a city name (no spaces, max length is 20 characters) and a unique identifier of the city (call it city Id.). You cannot assume anything about the city identifier, except that it is an integer that fits within 32-bit integer of C. You need to process the file to the end.
- **{max.txt, min.txt, avg.txt}**: Each line contains city id. Followed by temperature (real) information. max.txt contains the maximum temperature of each city, min.txt the minimum temperature; and avg.txt the average temperature. Each file should be read until the end. **No particular ordering can be assumed about the data.**

Your output is another file as described below:

- **cityTemps.txt**: This file should contain city temperature information. It should contain the cities in the same order as was in the cities.txt file. For each city, a line contains city name, followed by max, min and average temperatures respectively.

Write the necessary code. Then write necessary data files (using notepad) to test your code rigorously. Then ask your teacher to evaluate your assignment. A sample set of data files and corresponding output file is provided below

cities.txt	**max.txt**	**min.txt**	**avg.txt**
Dhaka 1552389 Chittagong 2323456 Rajshahi 5463782 Khulna 2484090	1552389 35.25 5463782 41.59 2484090 33.50 2323456 32.75	2484090 9.00 1552389 10.25 2323456 12.28 5463782 4.50	5463782 37.50 1552389 32.70 2323456 30.67 2484090 32.00

Output file: cityTemps.txt
Dhaka 35.25 10.25 32.70 Chittagong 32.75 12.28 30.67 Rajshahi 41.59 4.50 37.50 Khulna 33.50 9.00 32.00

For your convenience, a suitable structure definition is provided below:

```
struct City
{
      char name[20 + 1];  // Name of the city
      int id;             // Unique identifier of the city
      double maxT;        // maximum temperature
      double minT;        // minimum temperature
      double avgT;        // average temperature
} cities[100];
```

Binary mode file operations

Problem: ImageViewer

In this problem, you will implement an image viewer. The image file is a binary file with the following format:

- 1st Integer represents *width* of the image
- 2nd integer represents *height* of the image
- Then there are *width*height* integers.

Read these integers in the structure variable g_image in the given skeleton file. Write necessary code in the getImage() function and the image viewer should start working!

Random access file operations

Problem: ImageViewer V2

In this problem, the binary file will contain multiple images. Given an index, you should be able to load that image only and render it. The image file is a binary file with the following format:

- 1st integer *n* represents number of images in the file
- Next *n* integers represent the total number of bytes in each of the subsequent image data.
- For each image data that follows, the format is as before.

Use the concept of random access, so that when a specific number (0-9) is hit in the keyboard, the corresponding image is shown in the image viewer.